

## ZADUI, China

WMO: 560180

Lat: 32.8934N

Lon: 95.2949E

Elev: 13346

StdP: 8.86

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB  |             | Humidification DP/MCDB and HR |            |             |              |            |             | Coldest Month WS/MCDB |             |             |             | MCWS/PCWD to 99.6% DB |            | WSF          |
|---------------|-------------|-------------|-------------------------------|------------|-------------|--------------|------------|-------------|-----------------------|-------------|-------------|-------------|-----------------------|------------|--------------|
|               |             |             | 99.6%                         |            |             | 99%          |            |             | 0.4%                  |             | 1%          |             |                       |            |              |
|               | 99.6%       | 99%         | DP                            | HR         | MCDB        | DP           | HR         | MCDB        | WS                    | MCDB        | WS          | MCDB        | MCWS                  | PCWD       |              |
| (a)<br>1      | (b)<br>-6.0 | (c)<br>-2.0 | (d)<br>-25.6                  | (e)<br>2.2 | (f)<br>16.3 | (g)<br>-22.4 | (h)<br>2.6 | (i)<br>17.5 | (j)<br>23.3           | (k)<br>39.8 | (l)<br>20.7 | (m)<br>34.9 | (n)<br>2.2            | (o)<br>180 | (p)<br>0.701 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |             |             |             |             |             | Evaporation WB/MCDB |             |             |             |             |             | MCWS/PCWD to 0.4% DB |            |
|---------------|------------------------|-----------------|-------------|-------------|-------------|-------------|-------------|---------------------|-------------|-------------|-------------|-------------|-------------|----------------------|------------|
|               |                        | 0.4%            |             | 1%          |             | 2%          |             | 0.4%                |             | 1%          |             | 2%          |             |                      |            |
|               |                        | DB              | MCWB        | DB          | MCWB        | DB          | MCWB        | WB                  | MCDB        | WB          | MCDB        | WB          | MCDB        | MCWS                 | PCWD       |
| (a)<br>7      | (b)<br>19.1            | (c)<br>69.1     | (d)<br>49.0 | (e)<br>66.4 | (f)<br>48.5 | (g)<br>64.1 | (h)<br>47.6 | (i)<br>51.8         | (j)<br>64.0 | (k)<br>50.8 | (l)<br>62.3 | (m)<br>49.8 | (n)<br>60.6 | (o)<br>5.4           | (p)<br>110 |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 47.4                            | 80.8 | 55.8 | 46.5 | 78.0 | 55.2 | 45.6 | 75.4 | 54.6 | 27.2          | 64.3 | 26.4 | 62.6 | 25.6 | 61.0 |                   |

## Extreme Annual Design Conditions

| Extreme Annual WS |          |          | Extreme Annual Temperature |          |                    |          | n-Year Return Period Values of Extreme Temperature |           |            |           |            |           |            |           |    |    |
|-------------------|----------|----------|----------------------------|----------|--------------------|----------|----------------------------------------------------|-----------|------------|-----------|------------|-----------|------------|-----------|----|----|
|                   |          |          | Mean                       |          | Standard Deviation |          | n=5 years                                          |           | n=10 years |           | n=20 years |           | n=50 years |           |    |    |
| 1%                | 2.5%     | 5%       | Min                        | Max      | Min                | Max      | Min                                                | Max       | Min        | Max       | Min        | Max       | Min        | Max       | DB | WB |
| (a) 18.0          | (b) 14.5 | (c) 11.8 | (e) -12.7                  | (f) 74.1 | (g) 4.5            | (h) 2.5  | (i) -16.0                                          | (j) 75.9  | (k) -18.6  | (l) 77.4  | (m) -21.2  | (n) 78.8  | (o) -24.5  | (p) 80.6  |    |    |
| (4) 18.0          | (5) 14.5 | (6) 11.8 | (7) -13.8                  | (8) 54.2 | (9) 4.2            | (10) 1.4 | (11) -16.8                                         | (12) 55.2 | (13) -19.3 | (14) 56.0 | (15) -21.6 | (16) 56.7 | (17) -24.6 | (18) 57.7 |    |    |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 35.8   | 15.6  | 20.3  | 27.3  | 35.5  | 43.2  | 50.1  | 54.1  | 53.3  | 48.2  | 36.8  | 26.2  | 18.3  |
|                                                                                  | DBStd       | 14.39  | 6.84  | 6.27  | 5.27  | 4.65  | 4.35  | 4.46  | 3.45  | 3.63  | 4.28  | 5.52  | 4.70  | 6.31  |
|                                                                                  | HDD50       | 5514   | 1066  | 832   | 702   | 435   | 216   | 52    | 8     | 13    | 83    | 409   | 714   | 984   |
|                                                                                  | HDD65       | 10650  | 1531  | 1252  | 1167  | 885   | 675   | 447   | 338   | 363   | 505   | 874   | 1164  | 1449  |
|                                                                                  | CDD50       | 338    | 0     | 0     | 0     | 0     | 5     | 56    | 135   | 114   | 28    | 0     | 0     | 0     |
|                                                                                  | CDD65       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                                  | CDH74       | 5      | 0     | 0     | 0     | 0     | 0     | 1     | 2     | 2     | 0     | 0     | 0     | 0     |
|                                                                                  | CDH80       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
| Wind                                                                             | WSAvg       | 4.2    | 4.3   | 4.8   | 5.1   | 4.7   | 4.4   | 3.8   | 3.8   | 3.7   | 3.7   | 3.9   | 4.2   | 4.1   |
| Precipitation                                                                    | PrecAvg     | 20.9   | 0.2   | 0.2   | 0.4   | 0.6   | 1.7   | 4.6   | 4.5   | 3.9   | 3.3   | 1.2   | 0.2   | 0.2   |
|                                                                                  | PrecMax     | 27.3   | 1.0   | 0.5   | 1.0   | 1.4   | 3.7   | 7.7   | 8.5   | 6.2   | 6.5   | 5.1   | 0.7   | 0.9   |
|                                                                                  | PrecMin     | 16.2   | 0.0   | 0.0   | 0.0   | 0.1   | 0.3   | 1.9   | 1.9   | 1.6   | 1.5   | 0.0   | 0.0   | 0.0   |
|                                                                                  | PrecStd     | 2.9    | 0.2   | 0.2   | 0.2   | 0.4   | 0.7   | 1.1   | 1.5   | 1.1   | 1.2   | 1.0   | 0.2   | 0.2   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 44.6  | 44.7  | 52.8  | 59.9  | 68.3  | 69.8  | 72.6  | 73.0  | 67.3  | 61.6  | 47.7  | 43.8  |
|                                                                                  |             | MCWB   | 26.3  | 26.8  | 30.9  | 35.0  | 42.4  | 48.6  | 51.0  | 49.4  | 48.1  | 40.6  | 29.3  | 26.4  |
|                                                                                  | 2%          | DB     | 38.5  | 40.5  | 46.7  | 54.7  | 61.1  | 65.3  | 68.6  | 69.3  | 63.8  | 54.7  | 43.6  | 39.3  |
|                                                                                  |             | MCWB   | 23.3  | 24.8  | 28.4  | 33.6  | 40.1  | 47.6  | 50.0  | 49.1  | 46.9  | 38.6  | 27.5  | 23.9  |
|                                                                                  | 5%          | DB     | 33.2  | 37.2  | 43.1  | 51.0  | 57.4  | 62.4  | 66.1  | 66.3  | 60.8  | 50.7  | 40.8  | 35.3  |
|                                                                                  |             | MCWB   | 20.5  | 23.1  | 27.1  | 32.4  | 39.0  | 46.8  | 48.9  | 48.7  | 45.8  | 36.3  | 26.2  | 22.0  |
|                                                                                  | 10%         | DB     | 29.0  | 34.0  | 39.7  | 47.7  | 54.1  | 59.6  | 63.7  | 63.4  | 57.9  | 47.3  | 38.1  | 31.6  |
|                                                                                  |             | MCWB   | 18.6  | 21.6  | 25.8  | 31.2  | 37.8  | 46.0  | 48.4  | 48.1  | 44.7  | 35.0  | 25.1  | 20.0  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 27.0  | 27.3  | 31.7  | 37.6  | 46.1  | 51.8  | 53.7  | 53.2  | 50.4  | 43.7  | 30.8  | 27.3  |
|                                                                                  |             | MCDB   | 43.3  | 42.0  | 47.2  | 50.9  | 57.6  | 63.4  | 67.1  | 65.7  | 62.8  | 53.8  | 41.7  | 42.6  |
|                                                                                  | 2%          | WB     | 23.8  | 25.4  | 29.7  | 35.6  | 43.4  | 50.2  | 52.1  | 51.4  | 48.8  | 41.2  | 28.9  | 24.2  |
|                                                                                  |             | MCDB   | 36.8  | 38.4  | 43.0  | 49.0  | 55.0  | 61.3  | 64.6  | 63.5  | 60.4  | 50.9  | 40.3  | 38.3  |
|                                                                                  | 5%          | WB     | 21.3  | 23.8  | 28.2  | 34.1  | 41.7  | 48.8  | 51.1  | 50.4  | 47.5  | 39.1  | 27.4  | 22.2  |
|                                                                                  |             | MCDB   | 31.7  | 35.4  | 40.0  | 46.9  | 52.3  | 58.9  | 62.7  | 62.1  | 58.2  | 47.3  | 38.5  | 34.3  |
|                                                                                  | 10%         | WB     | 19.2  | 22.2  | 26.9  | 32.7  | 40.1  | 47.4  | 50.1  | 49.5  | 46.2  | 37.1  | 25.8  | 20.4  |
|                                                                                  |             | MCDB   | 28.2  | 32.6  | 37.8  | 44.5  | 49.9  | 56.7  | 60.7  | 60.3  | 55.5  | 44.3  | 36.5  | 30.8  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 24.7  | 23.8  | 22.7  | 21.5  | 19.6  | 17.6  | 19.1  | 19.8  | 19.5  | 19.8  | 24.0  | 26.0  |
|                                                                                  |             | MCDBR  | 31.8  | 30.3  | 28.7  | 28.4  | 26.9  | 22.3  | 24.1  | 26.0  | 24.1  | 24.8  | 28.1  | 32.0  |
|                                                                                  |             | MCWBR  | 20.2  | 18.4  | 15.1  | 12.4  | 10.7  | 8.4   | 8.8   | 9.4   | 9.7   | 10.8  | 15.3  | 19.8  |
|                                                                                  | 5% WB       | MCDBR  | 28.7  | 27.5  | 24.2  | 22.7  | 20.1  | 18.2  | 18.8  | 19.4  | 20.2  | 20.0  | 25.0  | 30.4  |
|                                                                                  |             | MCWBR  | 18.7  | 17.0  | 13.3  | 10.3  | 8.9   | 7.9   | 7.6   | 7.7   | 8.7   | 9.4   | 13.9  | 19.1  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        |        | 0.192 | 0.215 | 0.264 | 0.305 | 0.312 | 0.296 | 0.286 | 0.281 | 0.261 | 0.236 | 0.201 | 0.182 |
|                                                                                  | taud        |        | 2.597 | 2.523 | 2.383 | 2.294 | 2.362 | 2.521 | 2.607 | 2.610 | 2.636 | 2.627 | 2.639 | 2.678 |
|                                                                                  | Ebn at Noon |        | 340   | 338   | 324   | 312   | 308   | 312   | 314   | 316   | 321   | 325   | 334   | 341   |
|                                                                                  | Edh at Noon |        | 24    | 29    | 36    | 42    | 39    | 34    | 30    | 30    | 28    | 26    | 23    | 21    |
| All-Sky Solar<br>Radiation                                                       | RadAvg      |        | 1069  | 1341  | 1697  | 1982  | 2035  | 1857  | 1869  | 1721  | 1542  | 1356  | 1240  | 1055  |
|                                                                                  | RadStd      |        | 109   | 82    | 95    | 107   | 144   | 107   | 107   | 135   | 83    | 71    | 43    | 63    |

## Historical Trends

|                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                        |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| Station Only           | +0.88 | +2.14   | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| Regional (4 neighbors) | +0.81 | +1.31   | -2.72  | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page